

Harish Lal

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Experience

Investnet Asset Management Ltd

Aug 2023 – Present

Software Development Engineer 2

Jul 2026 – Present

- Led a cross-team OAuth 2.0 PKCE migration with the identity platform team (UMP), cutting SSO login time from 20–45s to 4–8s; root-caused a production auth bug (redirect URI breaking cookie state) that had eluded prior fixes and resolved it with a server-route change.
- Designed a scalable async export system (Hangfire background jobs + polling activity center) that decoupled 30s–10min report generation from the UI thread, eliminating a class of user-facing failures on long-running exports.
- Built an MCP server letting teammates query and update app config via natural language instead of an outdated, manual Razor UI, covering the ~70% of the app that is config-driven; in initial testing, adopted by 10 of 13 teammates on the team.

Software Development Engineer 1

Aug 2023 – Jul 2026

- Co-designed a schema-driven grid architecture that replaced 15 duplicate implementations with one configurable system, then scaled it to 122 widgets and page-level layouts in production, removing engineering as a bottleneck for new widgets.
- Shipped Custom Layout end to end (design, RBAC, firm-wide publish) after direct client requests, giving users a self-serve path that previously required an engineer.
- Received Star of the Quarter for ownership beyond level scope, including a from-spec rebuild of the legacy grouping panel (1000+ lines to under 400) using reusable hooks.

Projects

Graphly – Dependency-Graph Issue Tracker

Go, PostgreSQL, React, TypeScript, Docker

graphly.fly.dev | github.com/Lal-Jr/Graphly

- Built a full stack issue tracker where dependency links drive a scheduling engine: critical path, forecast dates, and 1,000-run Monte Carlo ship-date confidence.
- Designed for correctness under concurrent writes: cycle detection via recursive CTE behind row-level locking, compare-and-swap edits, and a transactional outbox with Postgres LISTEN/NOTIFY for real-time sync. Deployed to production on Fly.io.

ArchFlow – Distributed-System Simulator

Next.js, React, TypeScript

archflow-sim.vercel.app | github.com/Lal-Jr/ArchFlow

- Built a browser-based simulator with a from-scratch rate-based engine modeling queues, retries, circuit breakers, and autoscaling at 10 updates/sec, covering 24 system-design reference problems.
- Backed the engine and every reference design with 180 Vitest tests gated by CI; used `useSyncExternalStore` to keep live metrics from re-rendering the graph.

Circle of Life – Location-Based Community Platform

Go (Gin), PostGIS, Redis, Next.js

circleoflife-web.fly.dev | github.com/Lal-Jr/CircleOfLife

- Built a domain-driven Go backend with PostGIS radius queries and a distance/recency/type feed-ranking model; Redis caching, fixed-window rate limiting (Redis INCR/EXPIRE), and SSE for real-time updates.
- Secured the platform with JWT auth, bcrypt, and server-side XSS sanitization, and deployed it across Fly.io, Supabase, and Upstash.

Skills

Languages: TypeScript, JavaScript, C#, Go, SQL

Frontend: React, Next.js, Redux, React Query, Tailwind CSS, HTML, CSS

Backend: Go (Gin), Node.js, REST APIs, WebSockets/SSE, .NET, Hangfire, OAuth 2.0/PKCE, JWT, MCP

Data: PostgreSQL, PostGIS, Redis, schema design, indexing

Testing & DevOps: TDD, Vitest, Docker, GitHub Actions CI/CD, Git

Cloud: Fly.io, Supabase, Vercel; AWS (foundational: S3, Lambda, DynamoDB)

Education

Government Engineering College, Barton Hill

Aug 2019 – Jun 2023

B.Tech, Electrical Engineering